

A Lindbladian for holographic Brownian motion

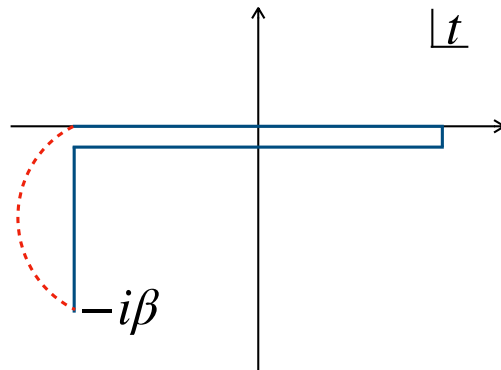
Daichi Takeda (RIKEN, iTHEMS)

June 27th, 2026

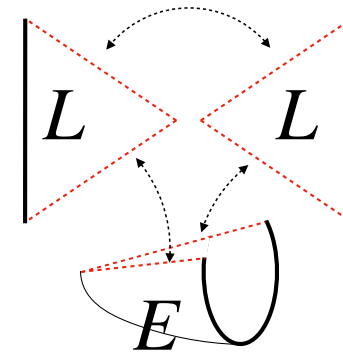
@ Holo-Asia 2026

Open quantum systems in AdS/CFT

Foundation: Schwinger-Keldysh effective action

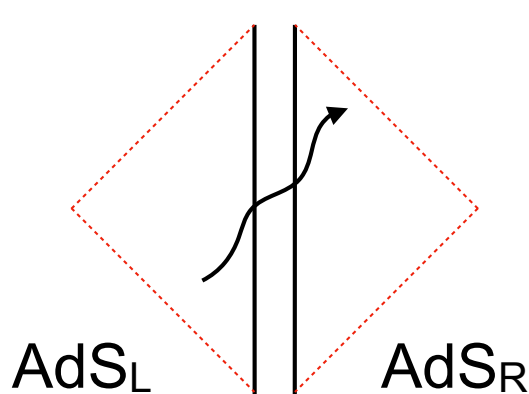


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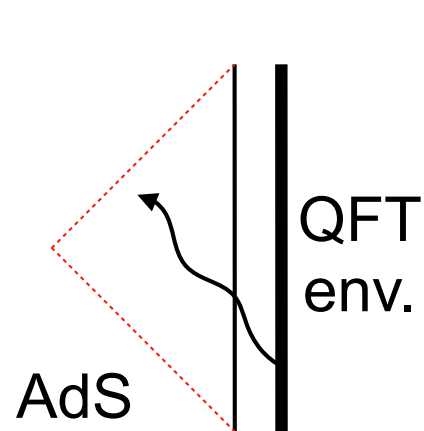


Finally established by
Skenderis, van Rees (2008)

Open holographic CFT



Aharony, Clark, Karch (2006)
and many successive works



Integrate out QFT first

Markovian case

→ Lindblad

Ishii, Takeda (2025)

non-Markovian case

Adachi, Hasegawa,
Ishii, Takeda (2026)

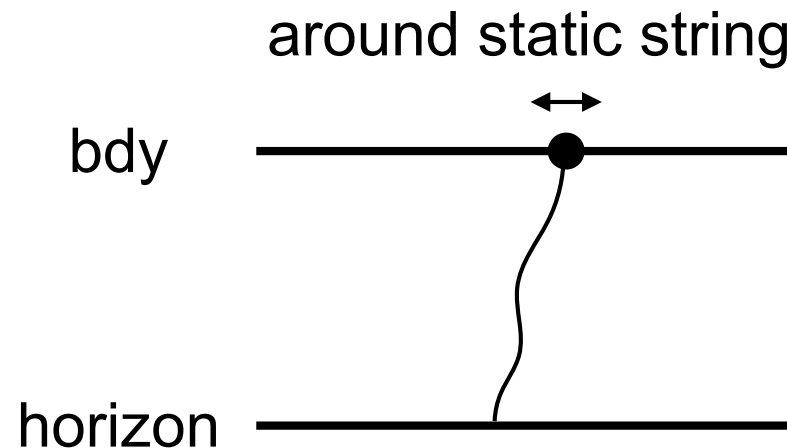
Heavy quark: use holographic CFT as a bath

Heavy probe quark in AdS/CFT

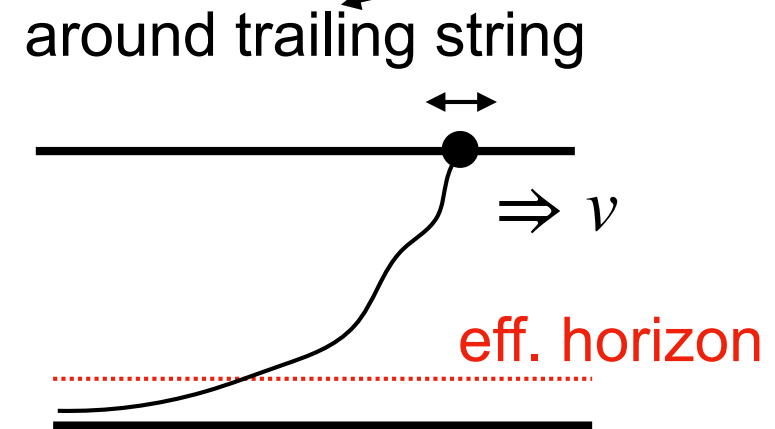
$$\left\langle \text{Tr} \exp \left(i \oint_C A_\mu dx^\mu \right) \right\rangle_{\text{CFT}} = \exp \left(i S_{\text{NG}}[X, \Sigma = \partial C] \right)$$

Holographic Brownian (Langevin) motion

Herzog, Karch, Kovtun,
Kozcaz, Yaffe (2006)
Gubser (2006)



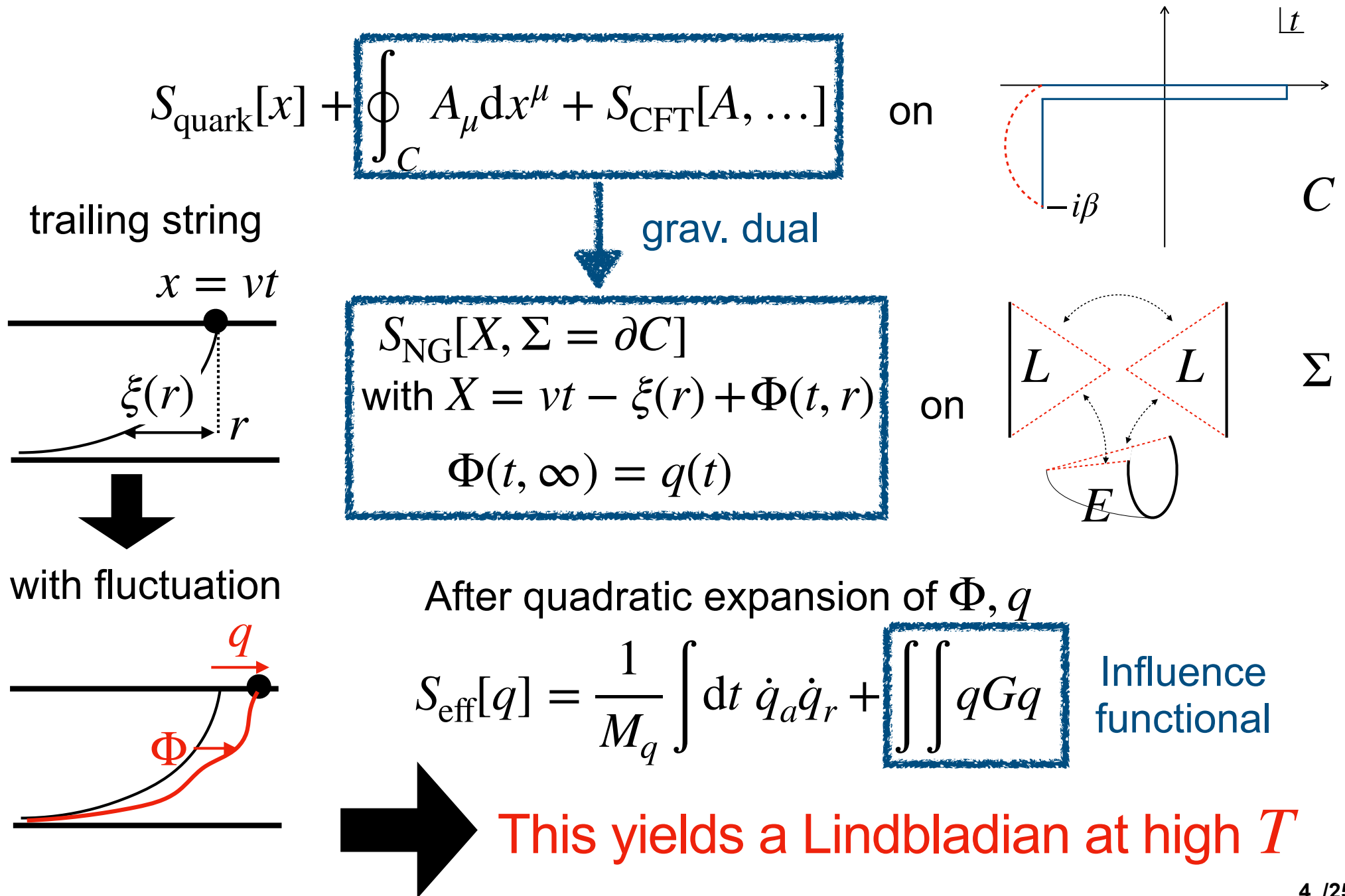
de Boer, Hubeny, Rangamani, Shigemori (2008)
Son, Teaney (2009)



Giecold, Iancu, Mueller (2009)
Casalderrey-Solana, Kim, Teaney (2009)

Can HBM be quantized?

HBM follows a Lindblad eq. at high T



Holographic QBM obeys a Lindblad master equation (in high T)

1. Influence functional & derivative expansion
→ a Lindbladian (lesson from Caldeira-Leggett)
2. Holographic BTZ bath gives a Lindbladian
3. Dynamics of the Lindbladian
4. Extension to AdS_5 black brane bath

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Open system by tracing out the bath

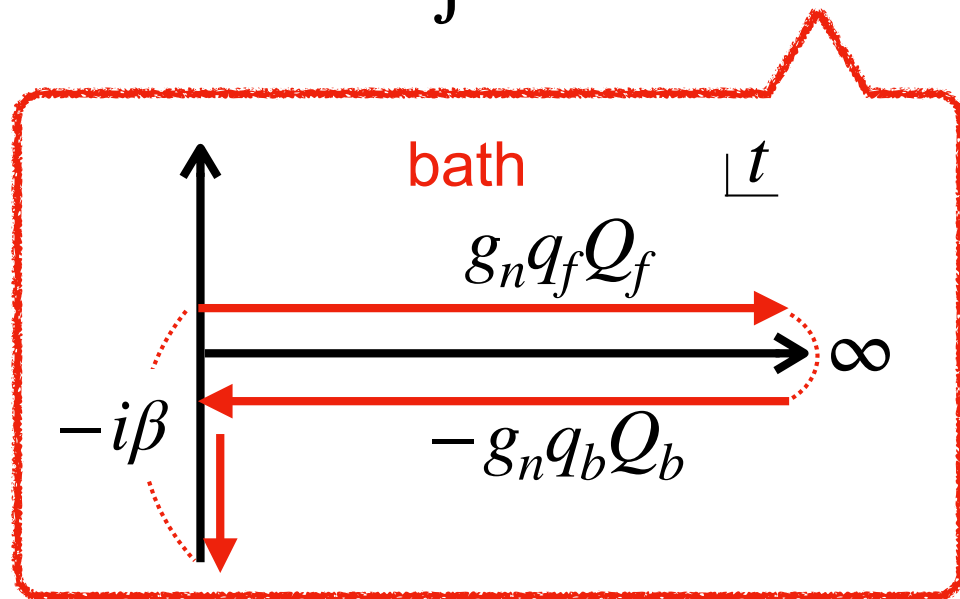
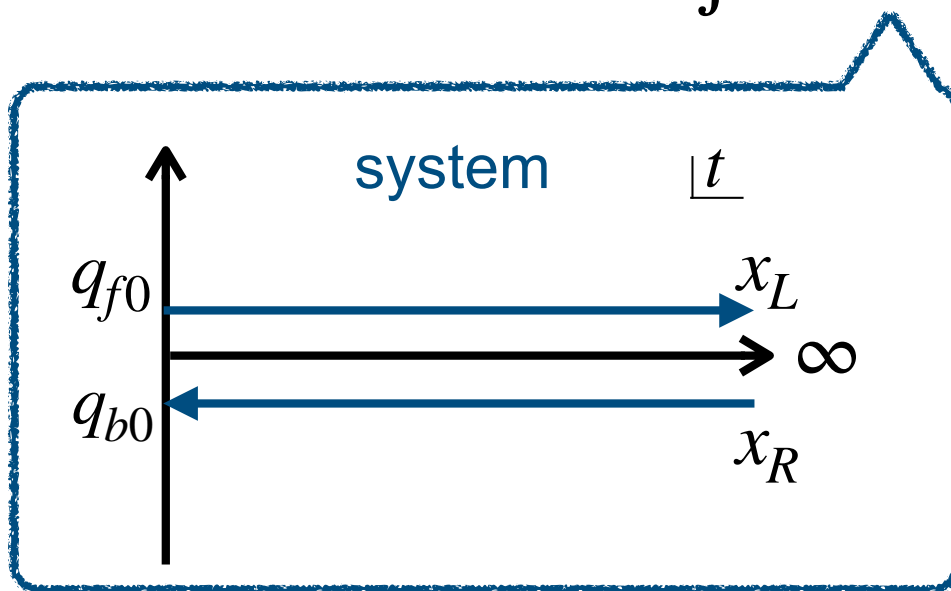
Caldeira-Leggett model

$$L = \frac{m}{2} \dot{q}^2 - V(q) + \sum_n \left[\frac{M_n}{2} \dot{Q}_n^2 - \frac{M_n \omega_n^2}{2} Q_n^2 + g_n Q_n q \right]$$

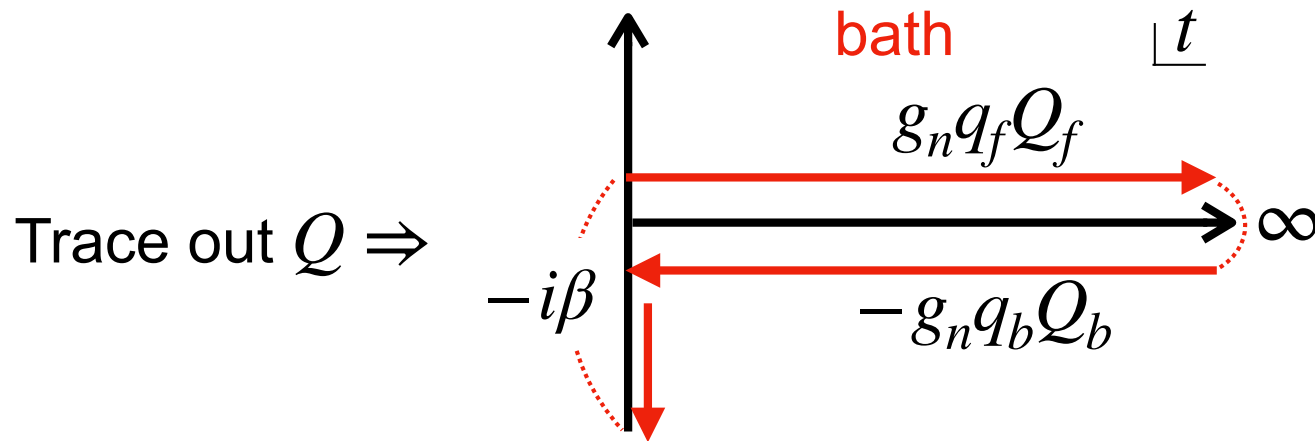
$$\text{with } \rho_{\text{tot}}(0) = \rho_0 \otimes \prod_n \frac{e^{-\beta H_n}}{Z_n}$$

Reduced density matrix

$$\langle x_L | \text{Tr}_B \rho_{\text{tot}}(t) | x_R \rangle = \int \mathcal{D}q e^{iS[q]} \langle q_{f0} | \rho_0 | q_{b0} \rangle \oint \mathcal{D}Q e^{iS_B[Q] + iS_{\text{int}}[q, Q]}$$



Influence functional $\ni$ bath Green functions



$$q_r := \frac{q_f + q_b}{2}$$

$$q_a := q_f - q_b$$

$$\rho_{\text{sys}}(t; x_L, x_R) = \int \mathcal{D}q \, e^{iS[q_f] - iS[q_b]} \langle q_{f0} | \rho_0 | q_{b0} \rangle \times$$

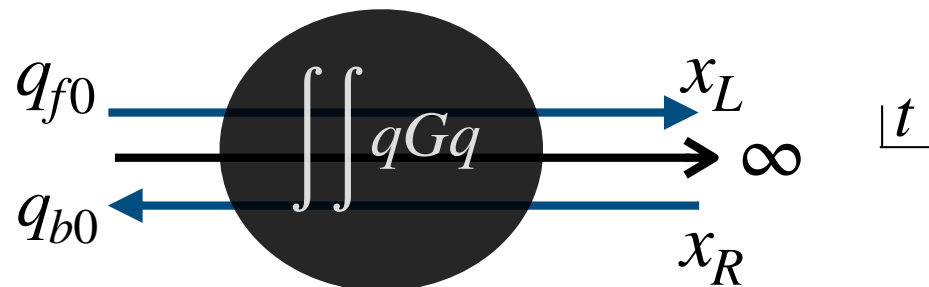
Influence functional

$$\exp \left[- \int dt dt' \left(\underbrace{2i q_a(t) \eta(t-t') \Theta(t-t') q_r(t')}_{\text{bath retarded Green func. (dissipation)}} + \frac{1}{2} q_a(t) \underbrace{\nu(t-t') q_a(t')}_{\text{bath symmetric Green func. (fluctuation)}} \right) \right]$$

bath retarded Green func.
(dissipation)

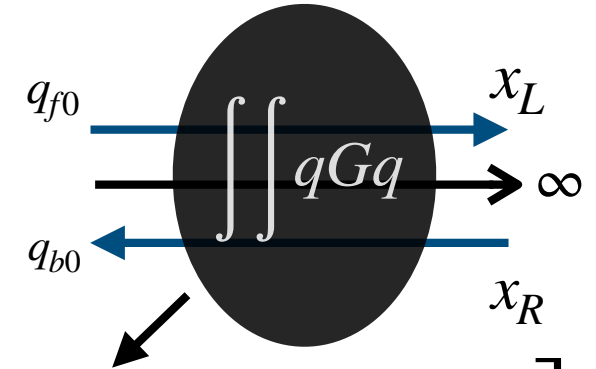
bath symmetric Green func.
(fluctuation)

system effective theory



Derivative expansion → Markovian

$$\rho_{\text{sys}}(t; x_L, x_R) = \int \mathcal{D}q \, e^{iS[q_f] - iS[q_b]} \langle q_{f0} | \rho_0 | q_{b0} \rangle \times$$



$$\exp \left[- \int dt dt' \left(2i q_a(t) \eta(t-t') \Theta(t-t') q_r(t') + \frac{1}{2} q_a(t) \nu(t-t') q_a(t') \right) \right]$$

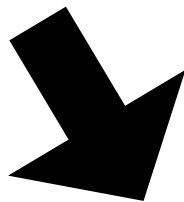
In the interaction picture

$$\rho_{\text{sys}}(t) = \text{T exp} \left[- \iint q G q \right] \rho_0$$

Derivative expansion at high T

$$\eta(t) \simeq \gamma \delta'(t)$$

$$\nu(t) \simeq 2\gamma T \delta(t) + (\gamma/6T) \delta''(t)$$

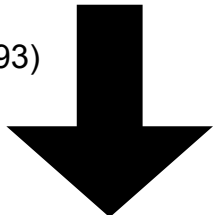


up to surface terms...

$$\rho_{\text{sys}}(t) \simeq \text{T exp} \left[- \int dt \left(-\gamma T q_a^2 - i\gamma q_a \dot{q}_r - \frac{\gamma}{12T} \dot{q}_a^2 \right) \right] \rho_0$$

Lindbladian (CPTP master eq)

$$\rho_{\text{sys}}(t) \simeq \text{T exp} \left[- \int dt \left(-\gamma T q_a^2 - i\gamma q_a \dot{q}_r - \frac{\gamma}{12T} \dot{q}_a^2 \right) \right] \rho_0$$

Diósi (1993)  t -derivative with $q_a \rho = (q_f - q_b) \rho = [q, \rho]$,
 $q_r \rho = \frac{1}{2} \{q, \rho\}$, $\dot{q} = p/m$

Master eq (in the interaction picture)

$$\dot{\rho} = -\gamma T [q, [q, \rho]] - i \frac{\gamma}{2m} [q, \{p, \rho\}] - \frac{\gamma}{12Tm^2} [p, [p, \rho]]$$

$$= \underbrace{-i \frac{\gamma}{2m} [qp, \rho]}_{\text{unitary}} + \underbrace{i C_{ij} \left(L_i \rho L_j^\dagger - \frac{1}{2} \{L_j^\dagger L_i, \rho\} \right)}_{\text{dissipation}} \quad L = \begin{pmatrix} q \\ p \end{pmatrix}$$

$$C = \begin{pmatrix} 2\gamma T & -i\gamma/2m \\ i\gamma/2m & \gamma/6Tm^2 \end{pmatrix} \begin{pmatrix} & \\ & \end{pmatrix} > 0 \Leftrightarrow \text{CPTP}$$

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When bath is holographic: BTZ case

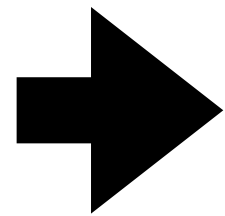
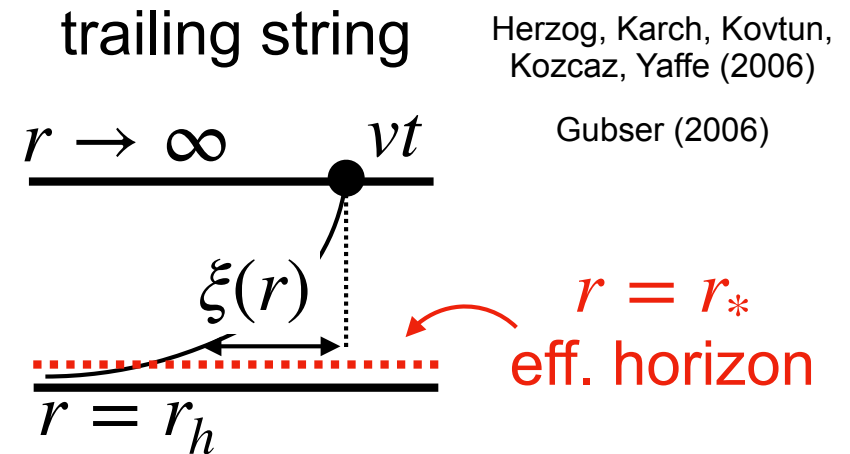
$$S_{\text{quark}}[x] + \int A_\mu dx^\mu + S_{\text{CFT}}[A, \dots] \quad \text{probe} \quad S_{\text{NG}}[X] \quad \text{on BTZ}$$

dual

zeroth order sol: $X = vt - \xi(r)$

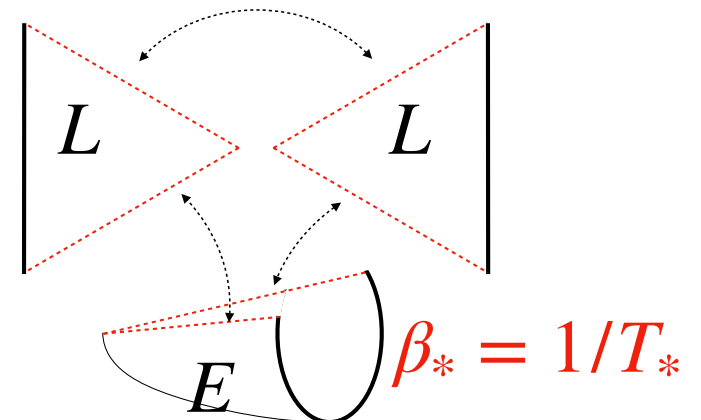
$$ds_{\text{WS}}^2 \sim -a(r - r_*)d\hat{t}^2 + b \frac{dr^2}{r - r_*}$$

($\hat{t}$: a WS time coordinate)

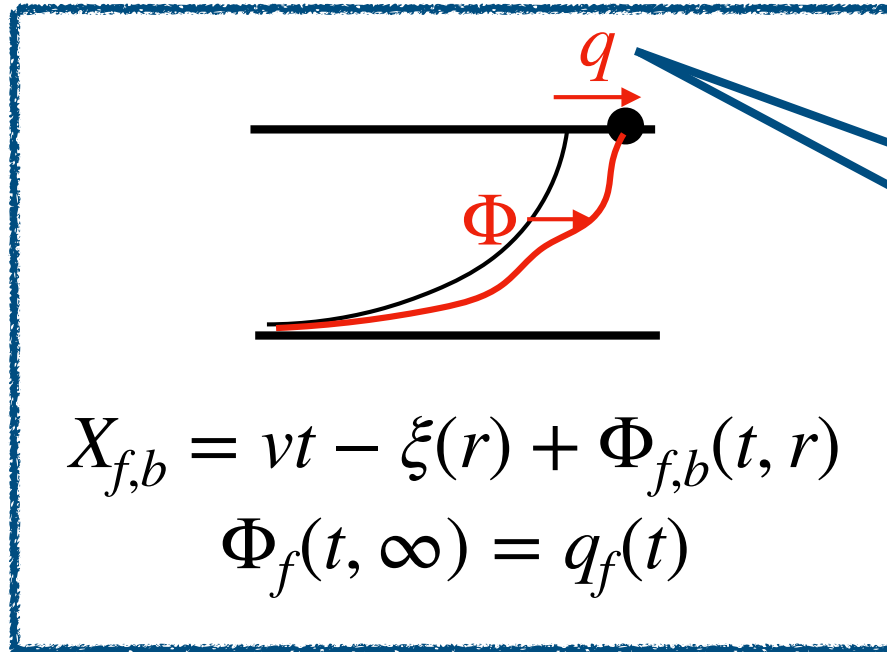


Consider a bulk SK contour with

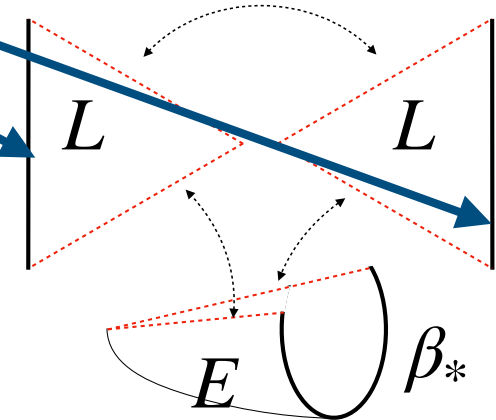
eff. temp. $T_* = \sqrt{1 + v^2} T_{\text{BTZ}}$



Add fluctuation



WS
Schwinger
-Keldysh



Up to const. and surface terms,

$$S_{\text{NG}}[X] \simeq -\frac{1}{2} \int d^2\sigma K(r) \sqrt{-h_0} h_0^{ab} \partial_a \Phi \partial_b \Phi$$

$$K(r) \propto r^2 + \text{const}$$

h_0 : zeroth order WS metric

Solve this Φ theory on the WS Schwinger-Keldysh

= Integrate out the bath CFT

Influence functional from WS Schwinger–Keldysh

On-shell action = Influence functional

(The exact solution of Φ is found)

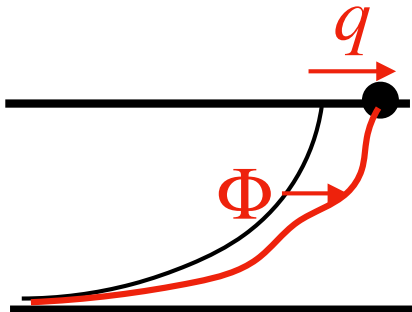
$$\mathcal{O}(q^2) \nearrow e^{iS_{\text{NG}}} = \exp \left[- \iint \left(i q_a G_R q_r + \frac{1}{2} q_a G_S q_a \right) \right]$$

Bu-Zhang (2023)
Nakamura-Takeda (2026)

The Green functions in the Fourier space

$$\tilde{G}_R(\omega) = -\frac{i}{2\pi\alpha'} \left[\frac{r_*^2(1+v^2)}{L^2(1-v^2)}\omega + \frac{L^2}{(1-v^2)^2}\omega^3 \right]$$

$$\tilde{G}_S(\omega) = -\text{Im } \tilde{G}_R(\omega) \coth \frac{\omega}{2T_*} \quad \text{*After counterterm subtraction}$$



Also, expand the quark action

$$S_{\text{quark}}[x] = -m \int ds \simeq \int dt \frac{M_q}{2} \dot{q}^2 \quad (M_q = \frac{m}{(1-v^2)^{3/2}})$$

Derivative expansion → Lindbladian

Ref. Hamiltonian

$$H_0 = \frac{p^2}{2M_q}$$

Interaction picture

$$\rho_{\text{sys}}(t) = \text{T exp} \left[- \int \int q G q \right] \rho_0$$

Derivative expansion of the Green functions

ω/T_* expansion

$$\tilde{G}_R(\omega) \simeq -i\gamma\omega$$

$$\tilde{G}_S(\omega) \simeq 2D_{pp} + 2D_{qq}M_q^2\omega^2$$

The Lindblad master eq (CPTP holds as before)

$$\dot{\rho} = -D_{pp}[q, [q, \rho]] - i\frac{\gamma}{2M_q}[q, \{p, \rho\}] - D_{qq}[p, [p, \rho]] \quad \text{Takeda (2026)}$$

	T_*	γ	D_{pp}	D_{qq}
BTZ	$\sqrt{1+v^2}T$	$\frac{2\pi L^2 T_*^2}{\alpha'(1-v^2)^2}$	γT_*	$\frac{\gamma}{12T_* M_q^2} \left(1 + \frac{3}{\pi^2}\right)$

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Boltzmann state is a steady state

Takeda (2026)

The master eq. in Schrödinger picture

$$\dot{\rho} = -i\left[\frac{p^2}{2M_q}, \rho\right] - D_{pp}[q, [q, \rho]] - i\frac{\gamma}{2M_q}[q, \{p, \rho\}] - D_{qq}[p, [p, \rho]]$$

$$= H_0 \quad (\text{free Hamiltonian})$$

Find a steady state by assuming $\rho = \rho_{ss}(p)$

$$\Rightarrow \frac{\gamma}{M_q} \frac{d}{dp} [p\rho_{ss}(p)] + D_{pp}\rho_{ss}''(p) = 0$$

The Boltzmann distribution

$$\rho_{ss}(p) \propto \exp\left[-\frac{\gamma}{2M_q D_{pp}} p^2\right] = \exp\left[-\frac{p^2}{2M_q T_*}\right] = e^{-H_0/T_*}$$

First moments (mean values)

Takeda (2026)

The master eq. in Schrödinger picture

$$\dot{\rho} = -i\left[\frac{p^2}{2M_q}, \rho\right] - D_{pp}[q, [q, \rho]] - i\frac{\gamma}{2M_q}[q, \{p, \rho\}] - D_{qq}[p, [p, \rho]]$$

Take trace with q and p

$$\frac{d}{dt}\langle q \rangle = \frac{1}{M_p}\langle p \rangle, \quad \frac{d}{dt}\langle p \rangle = -\frac{\gamma}{M_p}\langle p \rangle$$

This is a classical damped motion!

$$\langle p \rangle_t = e^{-\gamma t/M_q}\langle p \rangle_0, \quad \langle q \rangle_t = \langle q \rangle_0 + \frac{1 - e^{-\gamma t/M_q}}{\gamma}\langle p \rangle_0$$

Second moments (covariance)

Takeda (2026)

The master eq. in Schrödinger picture

$$\dot{\rho} = -i\left[\frac{p^2}{2M_q}, \rho\right] - D_{pp}[q, [q, \rho]] - i\frac{\gamma}{2M_q}[q, \{p, \rho\}] - D_{qq}[p, [p, \rho]]$$

Similarly, take trace with q^2 , p^2 and $qp + pq$ and get a system of three 1st-order diff. eqs.

It can be solved, but let's see the late time behavior

$$\left\langle \frac{p^2}{2M_q} \right\rangle \rightarrow \frac{T_*}{2}$$

equipartition law

$$\langle q^2 \rangle \rightarrow 2Dt$$

$$D = \frac{T_*}{\gamma} \left[1 + \frac{\gamma D_{qq}}{T_*} \right]$$

Position keeps diffusing

$T_*/M_q \ll 1$

$\sim \frac{T_*}{\gamma}$

Classical Langevin

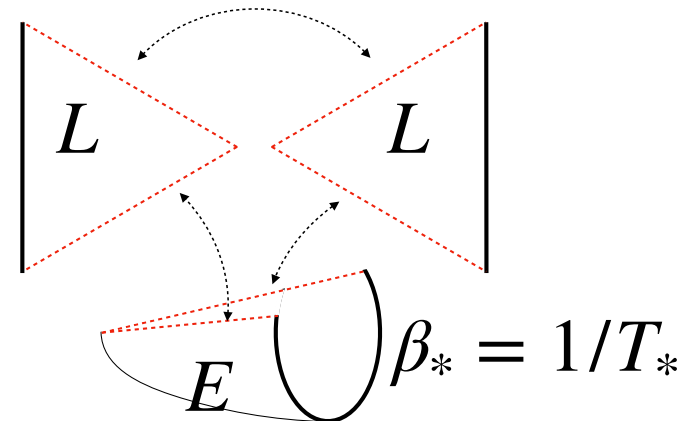
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Beyond BTZ: AdS₅ black brane

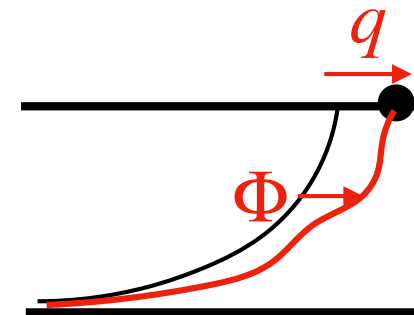
The formal structure is the same!

Consider a bulk SK contour with
 eff. temp. $T_* = (1 - v^2)^{1/4} T_{5D}$
 small change



Then, add longitudinal fluctuation and solve

$$S_{\text{NG}}[X] \simeq -\frac{1}{2} \int d^2\sigma K(r) \sqrt{-h_0} (\partial\Phi)^2$$



The time-evolution

$$\rho_{\text{sys}}(t) = \text{T exp} \left[- \iint q G q \right] \rho_0$$

No exact solution, but ω/T expansion is useful

The time-evolution

$$\rho_{\text{sys}}(t) = T \exp \left[- \int \int q G q \right] \rho_0$$

Difference: the exact G is not known
(ChatGPT also gave up)

However, ω/T_* -expansion is available!

Son, Starinets (2002)
Son, Teaney (2009)

For example, to get the ingoing solution with $\Phi|_{r \rightarrow \infty} \rightarrow 1$,

solve EOM by $\Phi_\omega = 1 + \omega \Phi_1 + \omega^2 \Phi_2 + \omega^3 \Phi_3 + \dots$
those are normalizable

together with the ingoing BC:

$$\Phi_\omega \sim \left(1 - \frac{r_*}{r} \right)^{-i\omega/4\pi T_*} (1 + \omega n_1 + \omega^2 n_2 + \dots)$$

Again, we get a Lindblad master eq!

Takeda (2026)

Ref. Hamiltonian

$$H_0 = \frac{p^2}{2M_q}$$

Interaction picture

$$\rho_{\text{sys}}(t) = \text{T exp} \left[- \int \int q G q \right] \rho_0$$

From the derivative expansion of the solution

$$\tilde{G}_R(\omega) \simeq -i\gamma\omega + \delta M \omega^2$$

Thermal mass
correction (new)

$$\tilde{G}_S(\omega) \simeq 2D_{pp} + 2D_{qq}M_q^2\omega^2$$

The master eq (in the interaction picture)

$$\dot{\rho} = -i \frac{\delta M}{M_q^2} [p^2, \rho] - D_{pp}[q, [q, \rho]] - i \frac{\gamma}{2M_q} [q, \{p, \rho\}] - D_{qq}[p, [p, \rho]]$$

mass correction

This is CPTP also for 5D coefficients

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Outlook

- Transverse fluctuation and 3D Lindbladian
- Rederive Lindbladian in bulk operator formalism
- Worldsheet causality
vs
irreversibility of Lindbladian